

From Questionnaire to Dataset

120 minutes · 6 self-check questions

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Turning completed forms into a rectangular dataset a computer can read: the codebook, variable naming, data entry, and the decisions that are expensive to reverse later.

Learning objectives

- Build a codebook that documents every variable and its coding
- Name and type variables so the dataset survives being handed to someone else
- Choose between wide and long layout for your design
- Set up double entry or electronic capture to control transcription error
- Import a dataset into SPSS and R and confirm it arrived intact

The codebook

- One row per variable: name, label, type, coding, permitted range, and the question it came from
- Write it before data entry begins, not afterwards — it is the specification, not the documentation
- Record the exact wording of the question, because "income" means nothing six months later
- Note the unit of measurement: kilograms or pounds, cedis or dollars, days or weeks
- Every recode you invent later gets added to the codebook, so the file always describes the data you actually have

IN PRACTICE

The codebook is what lets someone else — a supervisor, a co-author, a reviewer, or you in a year — understand the dataset without asking you. A study without one is a study only its author can analyse.

Structuring the dataset

- One row per unit of analysis, one column per variable, one value per cell
- The unit of analysis is usually the participant, but may be the visit, the household or the sample
- Wide format: one row per participant, repeated measures in separate columns (bp_baseline, bp_6months)
- Long format: one row per participant per occasion, with a time variable — required by most mixed-model and repeated-measures procedures

- Never store two pieces of information in one column ("35M" for age and sex)
- Never use colour or bold to encode meaning; a computer cannot read formatting

Naming and typing variables

- Short, lowercase, no spaces, no accents: age_years, sex, bp_systolic
- Do not start a name with a digit, and avoid names that are reserved words in R or SPSS
- Prefix grouped items consistently: phq_1 to phq_9, so they can be selected as a block
- Types: numeric for measurements, integer for counts, factor or categorical for groups, date for dates
- Coding categories as numbers with value labels is standard in SPSS; in R prefer factors with meaningful levels
- Reserve a consistent code for missing — and never use 0, 9 or 99 if those are plausible real values

A CODEBOOK EXTRACT

Variable	Label	Type	Coding / range
id	Participant ID	integer	1-600, unique
sex	Sex	categorical	1 = male, 2 = female
age_years	Age at interview	numeric	18-95
edu	Highest education	ordinal	1 = none, 2 = primary, 3 = JHS, 4 = SHS, 5 = tertiary
bp_systolic	Systolic BP (mmHg)	numeric	70-260
nhis	Has NHIS cover	binary	0 = no, 1 = yes
—	Missing value code	—	-99 across all variables

Getting the data in

- Electronic capture (KoBoToolbox, REDCap, ODK) enforces ranges and skip logic at the point of entry and removes transcription entirely
- On paper, use double entry: two people enter independently, then compare and reconcile differences
- Single entry with a 10% verification sample is the minimum defensible alternative
- Enter data exactly as recorded; correct implausible values as a documented cleaning step, not silently during entry
- Keep the raw file untouched and read-only — all cleaning happens in a script or a documented copy
- Back up before you start: three copies, two media, one off-site

Reading it into your software

- CSV is the safest interchange format; Excel silently reformats dates and long ID numbers
- In R: `dat <- read.csv("survey.csv", na.strings = "-99")`
- In SPSS: File > Import Data > CSV, then set measurement level and missing values in Variable View
- Immediately check dimensions, variable names and types against the codebook
- `str(dat)` in R, or Variable View in SPSS, will show a numeric variable that arrived as text
- A variable that should be numeric but imported as character almost always means a stray value — "N/A", a comment, or a thousands separator

Worked examples

The spreadsheet that cannot be analysed

A student arrives with 340 completed questionnaires typed into Excel. Ages are recorded as "35 yrs", "about 40" and "1987". Sex is coded M, F, m, f and Male. Missing answers are blank in some columns and "N/A" in others.

1. Nothing can be computed yet: age is text, so R will not take a mean and SPSS will treat it as nominal.
2. "1987" is a birth year, not an age — two different variables were entered into one column.
3. Five spellings of two categories means `table(sex)` returns five groups instead of two.
4. Blank and "N/A" both mean missing, but only blank will be recognised on import.
5. The fix is not cleverer code, it is a codebook and re-entry: age_years numeric, sex coded 1/2, one missing code throughout.
6. Re-entering 340 records takes two days. Untangling this by script takes longer and leaves doubt about every value.

THE POINT

Data entry decisions are analysis decisions. An hour spent on the codebook before entry saves days afterwards, and the days afterwards never fully restore confidence in the data.

Wide or long: the same data, two shapes

Blood pressure measured at baseline, 3 and 6 months for 120 patients. The student wants a repeated-measures analysis and the procedure refuses to run.

1. The data are wide: one row per patient with bp_o, bp_3, bp_6 as three columns.
2. Wide is convenient for computing change scores: `dat$change <- dat$bp_6 - dat$bp_o`.
3. Mixed models and most repeated-measures procedures need long: one row per patient per occasion.
4. In R: `long <- reshape(dat, direction = "long", varying = c("bp_o", "bp_3", "bp_6"), v.names = "bp", timevar = "month")`
5. Or with tidyr: `pivot_longer(dat, cols = starts_with("bp_"), names_to = "month", values_to = "bp")`
6. Keep both. They are the same data, and converting between them is two lines.

THE POINT

Neither shape is correct in general — the analysis decides. Knowing how to convert takes the argument away entirely.

Key terms

Codebook

A document defining every variable, its type, coding and permitted values.

Unit of analysis

The entity each row of the dataset represents — participant, visit, household.

Wide format

One row per participant, with repeated measures held in separate columns.

Long format

One row per participant per occasion, with an explicit time variable.

Double entry

Entering paper data twice independently and reconciling differences to catch transcription error.

Missing value code

A reserved value marking absent data, chosen so it cannot be confused with a real observation.

Activity

Take one page of your own questionnaire and write the codebook rows for every variable on it: name, label, type, coding and permitted range. Swap with a partner and try to enter three fictional respondents using only their codebook. Every question you have to ask them is a gap in it.

Further reading

- **Broman & Woo (2018)** — "Data Organization in Spreadsheets" — Short, free, and the best thing written on this. Read it before you enter any data.
- **Wickham (2014)** — "Tidy Data" — Why one variable per column matters, with examples of what goes wrong.
- **REDCap and KoBoToolbox documentation** — Both free for academic use; either removes a whole class of entry error.

Self-check questions

1. A column in your dataset contains entries like "35M" and "42F". The problem is that it:
 - a. Uses the wrong missing value code
 - b. Holds two variables in one column, so neither can be analysed
 - c. Is in long format
 - d. Should have been entered in Excel
2. Why is 99 a poor choice of missing value code for a variable recording age?
 - a. SPSS cannot handle two-digit codes
 - b. 99 is a plausible real age, so genuine and missing values become indistinguishable
 - c. It must always be negative
 - d. It conflicts with the participant ID
3. You measure blood pressure at baseline, 3 months and 6 months, and plan a repeated-measures analysis. The dataset should be in:
 - a. Wide format only
 - b. Long format, one row per participant per occasion
 - c. Whatever format it was entered in
 - d. Neither — repeated measures cannot be stored in a rectangle
4. A variable you know is numeric imports into R as character. The most likely cause is:

- a. The file was too large
 - b. A stray non-numeric entry somewhere in the column, such as "N/A" or a comment
 - c. R cannot read CSV files
 - d. The variable name starts with a letter
5. The strongest reason to keep the raw data file untouched is that:
- a. It saves disk space
 - b. Every cleaning decision stays reversible and auditable, because the original still exists
 - c. SPSS requires it
 - d. It makes the analysis run faster
6. When should the codebook be written?
- a. After analysis, as documentation
 - b. Before data entry begins, because it is the specification for the dataset
 - c. Only if the study is published
 - d. At the same time as the discussion chapter
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Answers

- 1. **b.** Age and sex are separate variables and need separate columns. As stored, you cannot compute a mean age or cross-tabulate by sex without first splitting the column — work that should never have been necessary.
 - 2. **b.** A missing code must be impossible as a real observation. For age, 99 is attainable; -99 is not. The same logic rules out 0 for variables where zero is a genuine value.
 - 3. **b.** Repeated-measures and mixed-model procedures expect long format: a row per observation with an explicit time variable. Wide is convenient for entry and for computing change scores, and converting between them is routine.
 - 4. **b.** A single non-numeric value forces the whole column to character. Find it with something like `unique(dat$var[is.na(as.numeric(dat$var))])` — it is usually a typed note, a unit, or a missing marker the import did not know about.
 - 5. **b.** Cleaning in place destroys the record of what the data originally said. Keeping the raw file read-only and doing all changes in a script means any decision can be revisited, and a reviewer can see exactly what you altered.
 - 6. **b.** The codebook defines what will be entered and how. Written afterwards it is archaeology, and it will not match the file, because entry decisions were made ad hoc while typing.
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Cleaning and Preparing Data

150 minutes · 6 self-check questions

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Finding what is wrong with a real dataset — impossible values, duplicates, inconsistent categories — and building the derived variables the analysis actually needs, all as a script you can rerun.

Learning objectives

- Screen every variable for implausible and out-of-range values
 - Detect and resolve duplicate records
 - Recode, collapse and reverse-score variables correctly
 - Compute derived variables such as BMI and scale totals
 - Decide whether an outlier is an error or a real observation
 - Keep cleaning reproducible by scripting it rather than editing cells
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First look at a new dataset

- Check dimensions against expectation: does the row count match the number of questionnaires?
 - Run a frequency table for every categorical variable and a min/max for every continuous one
 - In R: `summary(dat)` then `table(dat$sex, useNA = "ifany")` for each factor
 - In SPSS: Analyze > Descriptive Statistics > Frequencies, ticking Minimum and Maximum
 - Look for categories that should not exist — a sex coded 3, an education level of 7
 - Look for values at the boundaries: an age of 0, a blood pressure of 999
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Impossible, implausible and duplicate

- Impossible: outside the physically possible range — a diastolic BP of 400, a date of birth in the future
- Implausible: possible but unlikely enough to warrant checking — a 3-year-old with three children
- Cross-variable checks catch what single-variable checks miss: pregnant men, ages inconsistent with dates of birth
- Duplicates may be exact (entered twice) or partial (same participant, two IDs)
- Check for duplicates on the identifier and on a combination of key fields
- Resolve against the paper form where possible; where not, document the decision

IN PRACTICE

Return to the source before you "correct" anything. A value that looks impossible is sometimes a real and interesting observation, and sometimes a data entry slip — and only the original form tells you which.

Recoding and deriving

- Collapse categories only with a reason, and state it: small cell counts, or a clinically meaningful threshold
- Reverse-score negatively worded items before summing a scale, or the total is meaningless
- For a 1-5 item, the reversed value is 6 minus the original
- Compute scale totals only when the required number of items is present, and say how you handled partial scales
- Derived variables get their own names and codebook entries — never overwrite the source variable
- BMI in R: `dat$bmi <- dat$weight_kg / (dat$height_m)^2`
- Banding continuous variables loses information; do it for presentation, not for the primary analysis

Outliers

- An outlier is an observation far from the others; it is not automatically an error
- Detect with boxplots, standardised scores beyond about ± 3 , or simply sorting the variable
- Ask first whether it is plausible: a systolic BP of 210 is high but real; 2100 is a typing slip
- If it is an error and the true value is recoverable, correct it and document it
- If it is a genuine extreme value, keep it — and report a sensitivity analysis with and without it
- Deleting inconvenient observations because they weaken a result is data manipulation, not cleaning

Doing it reproducibly

- Every change goes in a script: R, SPSS syntax, or Stata do-file
- Point-and-click cleaning cannot be checked, cannot be rerun, and cannot be corrected when you find a mistake at step three
- Structure the script: read raw, clean, derive, save analysis dataset
- Never overwrite the raw file; write a separate cleaned file
- Keep a running count so you can report how many records were excluded and why
- Rerunning the whole script from raw data should reproduce the analysis dataset exactly

Worked examples

Six checks that catch most entry errors

A district survey of 600 households arrives as a clean-looking CSV. Before analysing anything, it needs screening.

1. Dimensions: `nrow(dat)` should be 600. It is 603 — three extra rows.
2. Duplicates: `sum(duplicated(dat$id))` returns 3. The same households were entered twice.
3. Ranges: `summary(dat$age_years)` shows a maximum of 199. A typing slip for 19 or 99.
4. Categories: `table(dat$edu)` shows a level 7 that the codebook does not define.
5. Cross-variable: `sum(dat$sex == 1 & dat$pregnant == 1)` returns 2 pregnant men.
6. Missing codes: `min(dat$income)` is -99, so the missing code was imported as a real value and every mean is wrong.

THE POINT

Every one of these passes a casual look at the spreadsheet. Six commands, run before any analysis, catch them all — and the last one would otherwise have silently corrupted every income statistic in the paper.

The outlier that was real

In a study of hypertension, one participant has a systolic pressure of 218 mmHg. It sits far above the rest and weakens the association the student expected. They ask whether to remove it.

1. First: is it possible? 218 mmHg is high but entirely attainable, and this is a hypertension study.
2. Second: check the source. The paper form says 218, recorded twice, with a note that the patient was referred.
3. So it is not an error. It is precisely the kind of participant the study exists to capture.
4. Removing it because it weakens the finding would be data manipulation, not cleaning.
5. Keep it, and run the analysis with and without it as a sensitivity check.
6. Report both: "excluding the participant with SBP 218, the association was 4.1 mmHg (95% CI 0.8 to 7.4) rather than 5.2".

THE POINT

The question is never "does this point help my result" but "is this value real". A sensitivity analysis lets you keep an honest observation and still show it is not driving the conclusion.

Key terms

Range check

Verifying every value of a variable falls within its permitted minimum and maximum.

Cross-variable check

Testing that combinations of values are logically consistent with each other.

Reverse scoring

Inverting negatively worded items so that all items on a scale point the same way.

Derived variable

A new variable computed from existing ones, such as BMI or a scale total.

Outlier

An observation distant from the rest of the distribution; an error or a genuine extreme.

Sensitivity analysis

Repeating an analysis under a different assumption to see whether the conclusion holds.

Activity

You are given a dataset of 600 records. Write down the checks you would run before analysing anything — list them variable by variable, including at least three cross-variable checks. Then write the R or SPSS syntax for two of them.

Further reading

- **Van den Broeck et al. (2005)** — "**Data Cleaning: Detecting, Diagnosing, and Editing Data Abnormalities**" — Open access, and the clearest framework: screening, diagnosis, treatment.
- **Osborne** — **Best Practices in Data Cleaning** — Practical, with worked examples of what goes wrong.
- **R for Data Science, chapters on data transformation** — Free online; dplyr makes cleaning scripts readable.

Self-check questions

1. A participant is recorded as male and 6 months pregnant. This is caught by:
 - a. A range check on sex
 - b. A cross-variable consistency check
 - c. A test for outliers
 - d. Cronbach's alpha
2. On a 1-5 agreement scale, the reversed value of a response of 2 is:

a. 3

- b. 4
 - c. 5
 - d. 2
3. A systolic blood pressure of 210 mmHg appears in your data. You should:
- a. Delete it as an outlier
 - b. Check it against the source, and if genuine keep it and consider a sensitivity analysis
 - c. Replace it with the mean
 - d. Round it down to the next plausible value
4. The strongest argument for scripting all cleaning rather than editing cells is that:
- a. Scripts run faster
 - b. The whole process can be rerun, checked and corrected, and the raw data stays intact
 - c. SPSS does not allow manual editing
 - d. It uses less memory
5. A total score is computed for participants who answered only 4 of 9 items. The safest approach is to:
- a. Sum whatever is present and treat it as a full score
 - b. Set a minimum number of completed items, apply it consistently, and report the rule
 - c. Always exclude anyone with any missing item
 - d. Replace missing items with zero
6. Which of these is data manipulation rather than cleaning?
- a. Correcting a transcription error against the paper form
 - b. Removing genuine observations because they weaken the association you expected
 - c. Flagging an impossible date of birth for checking
 - d. Documenting that three records were excluded as duplicates
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Answers

1. **b.** Both values are individually permissible — the impossibility only appears when they are considered together. Range checks alone would pass this record.
2. **b.** For a 1-k scale the reversed value is $(k + 1)$ minus the original, so $6 - 2 = 4$. Forgetting to reverse negatively worded items is one of the most common scale errors and quietly destroys the total.
3. **b.** It is high but entirely possible — this is exactly the kind of participant a study of hypertension exists to capture. Verify it, keep it, and show that the conclusion does not depend on it.
4. **b.** Manual edits are invisible and unrepeatable. If you discover an error in step three, a script lets you fix it and rerun; hand-edited cells mean starting over with no record of what was changed.

5. **b.** Summing four items gives a total that is not comparable to a nine-item total. Common practice is to require a stated proportion complete, prorate or impute the remainder, and say exactly what was done.
6. **b.** The distinction is whether the decision is driven by the data being wrong or by the result being unwelcome. Excluding valid observations to improve a finding is misconduct, however it is described in the write-up.



Missing Data

120 minutes · 6 self-check questions

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Why data are missing determines what you may do about it. Deletion, single imputation and multiple imputation, and what each assumes.

Learning objectives

- Quantify and display the pattern of missingness in a dataset
 - Distinguish MCAR, MAR and MNAR and explain why the distinction matters
 - State the cost of listwise and pairwise deletion
 - Explain why mean substitution understates uncertainty
 - Describe what multiple imputation does and when it is appropriate
 - Report missing data handling so a reader can judge it
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Look before you decide

- Report the amount missing per variable, not just overall
 - Look at the pattern: is missingness concentrated in a few participants, or spread thinly?
 - A variable missing for 40% of participants is a different problem from 40 variables each missing 1%
 - In R: `colMeans(is.na(dat))` gives the proportion missing per variable
 - Compare respondents and non-respondents on the variables you do have — this is the evidence for whether missingness is random
 - Item non-response (a skipped question) and unit non-response (a participant who never took part) are different problems
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Three mechanisms

- MCAR — Missing Completely At Random: missingness unrelated to anything, observed or not. A dropped test tube.
- MAR — Missing At Random: missingness explained by variables you observed. Older participants skip the income question more often, and you recorded age.
- MNAR — Missing Not At Random: missingness depends on the unobserved value itself. The highest earners decline to state income.
- MCAR is testable in part; MAR is an assumption; MNAR cannot be verified from the data at hand

- Most real missingness is MAR at best, which is precisely the case imputation is built for
- MNAR requires sensitivity analysis under different assumptions rather than a single fix

WHAT EACH MECHANISM PERMITS

Mechanism	Example	Deletion gives	Preferred handling
MCAR	Sample lost in transit	Unbiased, less precise	Complete cases acceptable
MAR	Older people skip income	Biased	Multiple imputation
MNAR	High earners refuse income	Biased	Sensitivity analysis; model the mechanism

Deletion

- Listwise (complete case) deletion drops any participant missing any analysis variable — the default in most software
- With 10 variables each 5% missing, complete cases can lose 40% of the sample
- It is unbiased only under MCAR; under MAR it biases estimates
- Pairwise deletion uses all available data for each computation, so different coefficients rest on different subsamples
- Pairwise can produce a correlation matrix that is mathematically impossible, which breaks regression and factor analysis
- Deletion is defensible when missingness is genuinely tiny — say under 5% and plausibly MCAR — and you say so

IN PRACTICE

Report how many participants each analysis actually used. A table saying $n = 600$ in the methods and a regression silently fitted on 380 is one of the easiest things for a reviewer to catch.

Imputation

- Mean substitution fills gaps with the variable mean; it preserves the mean, shrinks the variance, and biases standard errors downwards
- Because it pretends imputed values are as certain as observed ones, it manufactures significance
- Regression imputation predicts the missing value from other variables — better, but still too certain
- Multiple imputation creates several completed datasets, analyses each, and pools the results, so the uncertainty of imputation is carried into the standard errors
- Rubin's rules do the pooling; 5-20 imputations is typical, more if missingness is heavy
- In R: `mice(dat, m = 20) then pool(with(imp, lm(y ~ x)))`

- Include the outcome and any variable predictive of missingness in the imputation model, or you bias towards the null
- Last observation carried forward is not recommended for longitudinal data — it assumes no change, which is rarely true

Reporting

- State the amount missing per variable
- State the assumed mechanism and the evidence for it
- State the method used and the number of imputations if applicable
- State the analysis sample size for every model reported
- Where missingness is substantial, present a complete-case analysis alongside as a sensitivity check
- A reader should be able to judge whether your conclusion could plausibly be an artefact of the missing data

Worked examples

How 5% missing becomes 40% lost

A regression uses 10 variables, each with about 5% missing values, roughly independently. The dataset has 600 participants. The output says $n = 358$.

1. SPSS and R both delete any case missing any variable in the model — listwise deletion, silently.
2. The chance a participant is complete on all ten is 0.95^{10} , about 0.60.
3. 600×0.60 is 360, which matches the 358 reported.
4. So 40% of the sample was discarded without a warning appearing anywhere.
5. Worse, if missingness relates to age or income, the remaining 358 are not a random subset — the estimates are biased, not just imprecise.
6. Multiple imputation with `mice(dat, m = 20)` uses all 600 and carries the imputation uncertainty into the standard errors.

THE POINT

Always report the analysis n for every model, and compare it with the recruited n . A gap that large should be explained, not left for a reviewer to find.

Why the outcome goes into the imputation model

A student imputes missing income values but deliberately leaves the outcome — depression score — out of the imputation model, reasoning that including it would be circular.

1. The instinct is understandable but the statistics say the opposite.
2. Imputation is not prediction of the truth; it is generating values with the right joint distribution.
3. Omitting the outcome imputes incomes that carry no relationship to depression.
4. Those imputed cases then dilute the real association, biasing the estimate towards the null.
5. Include the outcome, and any variable that predicts missingness, in the imputation model.
6. Include more variables in the imputation model than in the analysis model — that is the recommended practice, not a mistake.

THE POINT

Leaving the outcome out feels cautious and is actively harmful. It is one of the most common imputation errors, and it always pushes results towards finding nothing.

Key terms

MCAR

Missing Completely At Random — missingness unrelated to observed or unobserved values.

MAR

Missing At Random — missingness explained by observed variables.

MNAR

Missing Not At Random — missingness depends on the unobserved value itself.

Listwise deletion

Excluding any case with a missing value on any variable in the analysis.

Multiple imputation

Creating several plausible completed datasets and pooling the results, so imputation uncertainty is reflected in the standard errors.

Rubin's rules

The formulae for combining estimates and standard errors across multiply imputed datasets.

Activity

For your own dataset, tabulate the percentage missing for every variable. Pick the worst one and argue, from what you know about how the data were collected, whether it is plausibly MCAR, MAR or MNAR — and say what that implies for how you should handle it.

Further reading

- **Sterne et al. (2009)** — "Multiple imputation for missing data in epidemiological and clinical research" — BMJ, open access, and the standard practical reference.
- **van Buuren** — **Flexible Imputation of Missing Data** — Free online; author of the mice package.
- **Little & Rubin** — **Statistical Analysis with Missing Data** — The theoretical foundation, if you need it.

Self-check questions

1. Older participants are more likely to leave the income question blank, and you recorded everyone's age. This missingness is:
 - a. MCAR
 - b. MAR
 - c. MNAR
 - d. Not missing data at all
2. The people who decline to state income are those with the highest incomes. This is:
 - a. MCAR
 - b. MAR
 - c. MNAR
 - d. Listwise deletion
3. Replacing missing values with the variable mean primarily causes:
 - a. The mean to shift
 - b. The variance to shrink and standard errors to be too small, overstating significance
 - c. The sample size to fall
 - d. No problem, provided under 10% is missing
4. Ten variables each have 5% missing, roughly independently. Complete-case analysis will retain approximately:

- a. 95% of the sample
 - b. 60% of the sample
 - c. 50% of the sample
 - d. 5% of the sample
5. The essential advantage of multiple imputation over single imputation is that it:
- a. Produces a larger sample
 - b. Carries the uncertainty of the imputation into the standard errors
 - c. Requires no assumptions
 - d. Works even when data are MNAR
6. The outcome variable should be included in the imputation model because omitting it:
- a. Makes the imputation run more slowly
 - b. Biases associations towards the null by imputing predictors as unrelated to the outcome
 - c. Is forbidden by Rubin's rules
 - d. Has no effect
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Answers

- 1. **b.** Missingness depends on age, which you observed. That is MAR — and it is exactly the condition under which multiple imputation performs well, because the imputation model can use age.
 - 2. **c.** Missingness depends on the unobserved value itself. No amount of imputation from observed variables fixes this; it calls for sensitivity analysis under different assumptions about the missing values.
 - 3. **b.** Mean substitution leaves the mean unchanged but piles values at the centre, shrinking variance. Treating imputed values as if they were observed manufactures precision the data do not support.
 - 4. **b.** 0.95 to the power of 10 is about 0.60 . Small amounts of missingness across many variables compound, which is why complete-case analysis so often discards far more data than researchers expect.
 - 5. **b.** Multiple completed datasets, analysed separately and pooled, propagate the fact that the imputed values were estimates. Single imputation treats a guess as data.
 - 6. **b.** Leaving the outcome out imputes predictor values that carry no relationship to it, diluting the very association you are estimating. It feels like circularity but is the correct procedure.
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Describing Data and Building Table 1

120 minutes · 6 self-check questions

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Summarising variables honestly: which measure of centre and spread fits which distribution, and how to assemble the participant characteristics table that opens almost every quantitative paper.

Learning objectives

- Choose an appropriate summary for each level of measurement
 - Judge distribution shape and pick mean or median accordingly
 - Compute and interpret standard deviation, IQR and confidence intervals
 - Build a Table 1 of participant characteristics
 - Decide when a p-value belongs in Table 1 and when it does not
 - Produce the table in SPSS and in R
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Matching the summary to the variable

- Nominal: counts and percentages only — a mean blood group is meaningless
 - Ordinal: median and interquartile range; means of Likert items are common but contested
 - Continuous and roughly symmetric: mean and standard deviation
 - Continuous and skewed: median and interquartile range
 - Always report the denominator alongside a percentage — 60% of 5 is not 60% of 500
 - Report percentages to at most one decimal place; more implies precision you do not have
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Judging the distribution

- Look at a histogram before choosing a summary; do not rely on a test alone
- Right skew is common for income, length of stay, biomarker concentrations and cost
- A mean far from the median signals skew
- Formal normality tests (Shapiro-Wilk) are oversensitive in large samples and underpowered in small ones
- In R: `hist(dat$x)` and `boxplot(dat$x)`; in SPSS, Analyze > Descriptive Statistics > Explore
- The question is not "is it exactly normal" but "is a mean a fair description of this distribution"

IN PRACTICE

Reporting a mean and standard deviation for a badly skewed variable is not wrong arithmetic — it is a misleading description. A mean income of GHS 4,200 with an SD of GHS 9,000 tells the reader almost nothing useful.

Spread and uncertainty

- Standard deviation describes the spread of the observations
- Standard error describes the precision of the estimate, and shrinks as the sample grows
- A confidence interval is the interval estimate that goes with a point estimate: $\text{mean} \pm 1.96 \times \text{SE}$ for a large sample
- Reporting SE instead of SD to make a figure look tidier misrepresents variability
- Interquartile range is the 25th to 75th percentile, and is robust to outliers
- Say which you are reporting: "mean (SD)" and "median (IQR)" are not interchangeable

CHOOSING A SUMMARY

Variable	Shape	Report	Example
Age	Roughly symmetric	Mean (SD)	42.3 (11.8) years
Income	Right skewed	Median (IQR)	GHS 1,800 (900-3,400)
Sex	Nominal	n (%)	312 (52.0%) female
Education	Ordinal	n (%) by level	104 (17.3%) tertiary
Length of stay	Right skewed	Median (IQR)	4 (2-9) days

Table 1

- One row per characteristic, columns for the whole sample and for each comparison group
- Give the denominator for every column, and note where it differs because of missing data
- Order rows sensibly: demographics, then clinical, then exposure
- Include the unit of measurement in the row label, not in every cell
- In R: the `tableone` or `gtsummary` package builds this directly from the data frame
- In SPSS: Analyze > Descriptive Statistics > Explore, split by group, then assemble

Should Table 1 contain p-values?

- In a randomised trial, no: any baseline difference is by definition due to chance, so testing it answers a question nobody asked
- CONSORT explicitly discourages significance testing of baseline characteristics
- In an observational study, a p-value can flag imbalance worth adjusting for, but the decision to adjust should be driven by subject knowledge, not by $p < 0.05$
- Standardised differences are a better measure of imbalance and do not depend on sample size
- Whatever you report, describe the comparison in the table footnote

Worked examples

Mean or median: an income variable

Monthly household income in a sample of 600: mean GHS 4,200, median GHS 1,800, standard deviation GHS 9,100, maximum GHS 180,000.

1. The mean is more than twice the median, which signals strong right skew.
2. The standard deviation exceeds the mean, so mean $\pm$ SD runs negative — impossible for income.
3. A histogram confirms it: most households below GHS 3,000, a long tail of a few very high earners.
4. Reporting "mean GHS 4,200 (SD 9,100)" describes almost nobody in the sample.
5. Report median GHS 1,800 (IQR 900 to 3,400), which describes the typical household.
6. If a mean is needed for a total or a cost calculation, report it separately and say why.

THE POINT

The arithmetic is fine either way; the description is not. When mean and median disagree sharply, the median is almost always the honest summary.

The p-value column that should not be there

A randomised trial submits Table 1 with a p-value column comparing baseline characteristics between arms. One row shows $p = 0.04$ for age, and the student wants to adjust for it.

1. Randomisation guarantees that any baseline difference arose by chance.
2. So the p-value tests a hypothesis known to be true, and 1 in 20 rows will be "significant" by construction.
3. CONSORT explicitly discourages significance testing of baseline characteristics.
4. The real question is not whether the difference is significant but whether it is large enough to matter.
5. A standardised difference answers that and does not depend on sample size.
6. Adjusting for a prognostic variable is fine — but decide that in advance from subject knowledge, not from a baseline p-value.

THE POINT

In a randomised trial, a baseline p-value answers a question nobody asked. In an observational study it can flag imbalance, but the decision to adjust still belongs to the conceptual framework, not the output.

Key terms

Standard deviation

The typical distance of observations from their mean; describes spread in the data.

Standard error

The standard deviation of an estimate; describes precision, and shrinks as n grows.

Interquartile range

The 25th to 75th percentile — a spread measure robust to outliers.

Skew

Asymmetry in a distribution, pulling the mean away from the median.

Table 1

The participant characteristics table summarising the sample, overall and by group.

Standardised difference

A measure of group imbalance independent of sample size, used instead of a p-value.

Activity

Build Table 1 for your own dataset, or a supplied one: overall column plus one column per group. For every row, justify in one line why you chose mean or median. Then decide whether a p-value column belongs, and defend it.

Further reading

- **Altman & Bland — Statistics Notes series, BMJ** — Short, free, and unmatched on describing data properly.
- **Assel et al. (2019) — "Guidelines for Reporting of Statistics"** — Concrete rules for tables and decimal places.
- **gtsummary package vignette (R)** — Produces a publication-ready Table 1 in a few lines.

Self-check questions

1. Household income in your sample has a mean of GHS 4,200 and a median of GHS 1,800. You should report:
 - a. Mean and standard deviation
 - b. Median and interquartile range, because the distribution is clearly right skewed
 - c. Either, they are equivalent
 - d. The mode
2. Standard error differs from standard deviation in that standard error:
 - a. Is always larger
 - b. Describes the precision of an estimate and shrinks as the sample grows
 - c. Can only be used for normal distributions
 - d. Is the square of the standard deviation
3. A reviewer objects to p-values in Table 1 of your randomised trial. They are:
 - a. Wrong — testing baseline balance is standard
 - b. Right — any baseline difference in a randomised trial arose by chance, so the test answers no useful question
 - c. Right only if the trial is small
 - d. Wrong, provided the test is non-parametric
4. Reporting "34.6% of participants" when the group contains 26 people is poor practice because:

- a. Percentages should never be used
 - b. One decimal place implies precision the denominator cannot support, and the count should be shown
 - c. 34.6% is mathematically impossible
 - d. SPSS cannot compute it
5. A Shapiro-Wilk test on 3,000 observations returns $p < 0.001$. The most sensible response is:
- a. Immediately switch to non-parametric tests
 - b. Look at a histogram — in large samples the test detects trivial departures from normality
 - c. Transform the variable regardless of its shape
 - d. Delete the outliers until it passes
6. For an ordinal education variable with five levels, the appropriate summary is:
- a. Mean and standard deviation
 - b. Counts and percentages for each level
 - c. Standard error only
 - d. Correlation coefficient
-

Answers

- 1. **b.** A mean more than double the median indicates strong right skew, typical of income. The median describes the typical household; the mean is pulled up by a few high earners.
 - 2. **b.** SD describes variability among observations and does not systematically shrink with n . SE describes how precisely the mean is estimated and falls as the square root of n . Substituting SE for SD makes data look less variable than it is.
 - 3. **b.** Randomisation guarantees that baseline differences are chance. CONSORT discourages the practice; standardised differences are more informative if imbalance needs describing.
 - 4. **b.** With $n = 26$ each participant is 3.8 percentage points. Report "9 (35%)" so the reader sees what the percentage rests on.
 - 5. **b.** Normality tests are oversensitive in large samples: a departure too small to matter still returns a tiny p -value. Judge the shape visually and ask whether the mean is a fair description.
 - 6. **b.** The categories are ordered but the intervals between them are not equal, so a mean is not interpretable. Report n (%) per level, or a median category if a single summary is needed.
-

Visualising Data

120 minutes · 5 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Choosing a chart that shows what the data actually do, and avoiding the plots that hide the very thing a reader needs to see.

Learning objectives

- Match a chart type to the variables being shown
 - Read a histogram, boxplot and scatterplot for shape, spread and outliers
 - Explain why bar charts of means conceal distribution
 - Build publication-quality figures in R and SPSS
 - Apply the conventions that make a figure readable in print and to colour-blind readers
-

Which chart for which variables

- One continuous variable: histogram or density plot for shape; boxplot for summary and outliers
 - One categorical variable: bar chart of counts — never a pie chart with more than three slices
 - Continuous by categorical: boxplot, violin plot, or a dot plot showing every observation
 - Two continuous: scatterplot, with a fitted line only if you are claiming a relationship
 - Change over time: line plot with time on the horizontal axis
 - Two categorical: grouped or stacked bar chart, or a mosaic plot
-

The bar chart of means problem

- A bar with an error bar shows two numbers and hides the entire distribution
- Radically different datasets — bimodal, skewed, or containing outliers — produce identical bars
- It also implies the bar length is meaningful from zero, which is false for many measurements
- For group comparisons prefer a boxplot, or a dot plot with every point drawn
- With small n , plot every observation; there is no reason to summarise 12 points
- If you must use bars, state whether the error bar is SD, SE or a confidence interval — they differ by a factor of several

IN PRACTICE

Journals in several fields now refuse bar charts of continuous data for exactly this reason. Showing the points costs nothing and lets a reader see what you saw.

Making figures honest

- Start a bar chart axis at zero; truncating it exaggerates differences
- For line and scatter plots a non-zero axis is acceptable, but say so if the range is narrow
- Label both axes with the variable and its unit
- Do not use a second vertical axis to imply a relationship between two series
- Avoid three-dimensional effects entirely — they distort area and length
- Aspect ratio changes the apparent steepness of a slope; keep it consistent when comparing panels

Colour and accessibility

- Roughly 1 in 12 men has some form of colour vision deficiency, most commonly red-green
- Never use red and green as the only distinction between two series
- Use the viridis palette, or vary shape and line type as well as colour
- Check a figure by converting it to greyscale — if it still reads, it will survive printing
- Keep the palette consistent across every figure in a paper or deck
- Use colour to encode data, not for decoration

CHART CHOICE AT A GLANCE

You have	Show shape with	Compare groups with
One continuous variable	Histogram, density plot	Boxplot by group
One categorical variable	Bar chart of counts	Grouped bar chart
Two continuous variables	Scatterplot	Scatterplot coloured by group
Continuous over time	Line plot	One line per group
Small samples ($n < 20$)	Plot every point	Dot plot with median line

Doing it in software

- R base: `hist(dats$bp)`, `boxplot(bp ~ group, data = dat)`, `plot(x, y)`
- ggplot2 gives publication quality: `ggplot(dat, aes(group, bp)) + geom_boxplot() + geom_jitter(width = 0.15)`

- `geom_jitter` over a boxplot shows both the summary and the raw observations
- SPSS: Graphs > Chart Builder, then double-click the chart to edit elements
- Export at 300 dpi or higher, or as a vector format (PDF, EPS, SVG), never a screenshot
- Set the figure size at export rather than resizing afterwards, or the text will not scale with it

Worked examples

Four datasets, one bar chart

Three groups are compared on a continuous outcome. The bar chart of means with error bars looks clean and shows a clear difference. The reviewer asks to see the data.

1. Group A is tightly clustered around its mean.
2. Group B is bimodal — two subgroups, and almost nobody near the mean itself.
3. Group C is skewed with three extreme values pulling the mean up.
4. All three produce a nearly identical bar and error bar.
5. Replacing it with `ggplot(dat, aes(group, y)) + geom_boxplot() + geom_jitter(width = 0.15)` shows all of this at once.
6. The bimodality in group B turns out to be sex, which was the real finding.

THE POINT

A bar chart shows two numbers and hides everything else. Plotting the points costs nothing and sometimes hands you the actual result.

The axis that exaggerated a difference

A slide shows attendance rising from 78% to 82% after an intervention. The bars make it look like a doubling.

1. The vertical axis starts at 76, so the visible bar heights are 2 and 6 — a threefold difference.
2. Bar length encodes magnitude from zero, so the eye reads the ratio of the drawn heights.
3. Redrawn from zero, the bars are nearly the same height, which is the honest picture.
4. If the difference genuinely matters at that scale, a line or dot plot with a restricted axis is acceptable — position, not length, carries the meaning there.
5. Either way, show the confidence intervals: 78% (95% CI 74 to 82) and 82% (95% CI 78 to 86) overlap substantially.
6. The 4-point difference may be real, but the chart was claiming far more than the data support.

THE POINT

Truncating a bar axis is the most common way an honest researcher produces a misleading figure. If the effect only looks impressive on a cropped axis, it is not impressive.

Key terms

Histogram

A chart of the distribution of one continuous variable, using binned counts.

Boxplot

A summary showing median, quartiles and outliers, useful for comparing groups.

Jitter

Small random horizontal displacement so overlapping points become visible.

Truncated axis

An axis not starting at zero, which exaggerates apparent differences in a bar chart.

Viridis

A perceptually uniform colour palette that remains readable in greyscale and to colour-blind viewers.

Vector format

A figure stored as shapes rather than pixels, so it stays sharp at any size.

Activity

Take one finding from your own data and draw it three different ways — for example a bar of means, a boxplot, and a dot plot of every observation. Decide which one a sceptical reader would trust most, and say why the other two are less honest.

Further reading

- **Weissgerber et al. (2015) — "Beyond Bar and Line Graphs"** — PLOS Biology, open access. The paper that changed several journals' policies.
- **Wilke — Fundamentals of Data Visualization** — Free online, and the best single reference on chart choice.
- **ggplot2 documentation and the R Graph Gallery** — Copyable examples for almost any chart you need.

Self-check questions

1. The main criticism of a bar chart showing group means with error bars is that it:
 - a. Is difficult to produce in R
 - b. Hides the distribution, so very different datasets can produce identical bars
 - c. Cannot show more than two groups
 - d. Requires the data to be normally distributed
2. You are comparing a continuous outcome across three groups with 15 participants each. The best chart is:
 - a. A bar chart of means
 - b. A boxplot or dot plot showing every observation
 - c. A pie chart
 - d. A line plot
3. Using only red and green to distinguish two lines is a problem because:
 - a. Those colours print poorly
 - b. Red-green deficiency is the most common form of colour blindness, affecting roughly 1 in 12 men
 - c. Journals charge extra for colour
 - d. It violates copyright
4. A bar chart with a vertical axis starting at 45 instead of 0:

- a. Is standard practice for clarity
 - b. Exaggerates the apparent difference between bars
 - c. Is required when values are large
 - d. Has no effect on interpretation
5. Which R code overlays the raw observations on a boxplot?
- a. `geom_boxplot() + geom_bar()`
 - b. `geom_boxplot() + geom_jitter(width = 0.15)`
 - c. `geom_histogram() + geom_density()`
 - d. `geom_line() + geom_point()`
-

Answers

- 1. **b.** Two numbers cannot describe a distribution. Bimodal data, skewed data and data with an extreme outlier can all yield the same bar and error bar, which is why boxplots or plotting every point are preferred.
 - 2. **b.** With 15 per group there is no reason to hide the data. A boxplot with jittered points shows the summary and every observation at once.
 - 3. **b.** A substantial minority of readers cannot separate them. Vary line type or shape as well as colour, and check the figure in greyscale.
 - 4. **b.** Bar length encodes magnitude from zero. Truncating the axis makes a small difference look dramatic. For line and scatter plots a non-zero axis is acceptable, because position rather than length carries the meaning.
 - 5. **b.** `geom_jitter` displaces points slightly so overlapping observations become visible. Layered over a boxplot it gives both the summary and the underlying data.
-

The Logic of Inference

150 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

What a p-value is and is not, what a confidence interval actually tells you, and why an effect size belongs in every result you report.

Learning objectives

- Explain sampling variation and the sampling distribution
 - Interpret a confidence interval correctly, and reject the common misreadings
 - State precisely what a p-value measures
 - Distinguish statistical significance from practical importance
 - Calculate and interpret common effect sizes
 - Explain Type I and Type II error, power, and the cost of multiple testing
-

Why samples vary

- Any sample is one of many that could have been drawn; each gives a slightly different estimate
 - The sampling distribution is the distribution of an estimate across all possible samples
 - Its standard deviation is the standard error of the estimate
 - The central limit theorem says that for a large enough sample the sampling distribution of the mean is approximately normal, whatever the shape of the underlying data
 - This is why means can be handled with normal-theory methods even when the raw data are skewed
 - It says nothing about the data being normal — a persistent confusion
-

Confidence intervals

- A 95% confidence interval is constructed so that, over repeated sampling, 95% of such intervals contain the true value
- It is not "a 95% probability the true value lies in this interval" — the true value is fixed, the interval is what varies
- Its width reflects precision: wide means the study could not pin the effect down
- A CI that includes the null value corresponds to $p > 0.05$, but carries far more information
- Report intervals for every effect estimate, not just p-values

- A non-significant result with a narrow interval around zero is evidence of no meaningful effect; with a wide interval it is simply uninformative

IN PRACTICE

"No significant difference" and "no difference" are not the same claim. The confidence interval is what tells them apart, which is the single strongest argument for always reporting it.

What a p-value is

- The probability of observing data at least as extreme as yours, if the null hypothesis were true
- It is not the probability the null hypothesis is true
- It is not the probability your finding is a fluke
- It is not a measure of effect size — a tiny, unimportant difference gives a small p in a large enough sample
- 0.05 is a convention, not a law of nature, and nothing decisive happens between 0.049 and 0.051
- Report exact values ($p = 0.03$), not thresholds ($p < 0.05$), and never " $p = 0.000$ " — write $p < 0.001$

Effect size

- The effect size answers "how big", which is what a reader actually wants to know
- Cohen's d for a difference between two means: roughly 0.2 small, 0.5 medium, 0.8 large — as rough guides only
- Correlation r, or r-squared, for the strength of association
- Odds ratio, risk ratio and risk difference for binary outcomes
- Eta-squared or partial eta-squared for ANOVA
- Always interpret the effect in the units of the problem: "8 mmHg lower" means more to a clinician than " $d = 0.42$ "
- Statistical significance with a trivial effect size is a large-sample artefact, not a finding

WHAT EACH NUMBER TELLS YOU

Quantity	Answers	Depends on sample size?
p-value	Is this compatible with no effect?	Yes, strongly
Effect size	How big is the effect?	No
Confidence interval	How precisely do we know it?	Yes, width shrinks with n
Power	Could this study have detected it?	Yes

Errors, power and multiplicity

- Type I error: concluding there is an effect when there is none. Its rate is the significance level, conventionally 5%
- Type II error: missing a real effect. Its rate is beta; power is 1 minus beta
- 80% power is conventional, meaning a 1 in 5 chance of missing a real effect of the size you specified
- An underpowered study that finds nothing has not shown absence of an effect
- Testing 20 hypotheses at 5% gives roughly a 64% chance of at least one false positive
- Adjust for multiplicity (Bonferroni is simple and conservative; Benjamini-Hochberg controls false discovery rate), or pre-specify one primary outcome
- Pre-specifying the primary analysis is the cleanest defence against fishing

Worked examples

Two non-significant results that mean opposite things

Two trials of the same drug both report "no significant difference" ($p = 0.31$ and $p = 0.28$). A reviewer treats them as equivalent evidence that the drug does not work.

1. Trial A: difference 0.4 mmHg, 95% CI -0.8 to 1.6. Narrow, centred near zero.
2. Trial B: difference 6.0 mmHg, 95% CI -5.5 to 17.5. Wide, spanning clinically important effects in both directions.
3. Trial A has effectively excluded a meaningful effect — its interval contains nothing that would change practice.
4. Trial B is simply uninformative; it is compatible with the drug being useless or highly effective.
5. The p-values are almost identical and tell you none of this.
6. Reporting only " $p > 0.05$ " would make the two studies indistinguishable in a review.

THE POINT

Absence of significance is not evidence of absence. The confidence interval is what separates "we showed there is no meaningful effect" from "we could not tell", and it is why intervals belong in every result.

Twenty tests, one press release

A study measures 20 outcomes, finds one significant at $p = 0.04$, and the abstract reports it as the main finding.

1. With 20 independent tests at the 5% level, the chance of at least one false positive is $1 - 0.95^{20}$, about 64%.
2. So finding one significant result among twenty is the expected outcome under no effect at all.
3. A Bonferroni-adjusted threshold would be $0.05/20 = 0.0025$; $p = 0.04$ does not come close.
4. Benjamini-Hochberg is less conservative and controls the false discovery rate instead.
5. The cleaner protection is to pre-specify one primary outcome in the protocol, before seeing data.
6. The finding is not worthless — it is hypothesis-generating, and should be reported as exploratory.

THE POINT

Testing many things and reporting the winner inverts the meaning of the p-value. Pre-specifying the primary outcome is the cheapest defence, and it has to happen before the analysis, not after.

Key terms

Sampling distribution

The distribution of an estimate across all possible samples of a given size.

Standard error

The standard deviation of the sampling distribution; the precision of an estimate.

Confidence interval

A range constructed so that a stated proportion of such intervals contain the true value over repeated sampling.

p-value

The probability of data at least as extreme as observed, assuming the null hypothesis is true.

Effect size

A measure of the magnitude of an effect, independent of sample size.

Power

The probability of detecting an effect of a specified size when it genuinely exists.

Multiplicity

The inflation of false positive risk that comes from testing many hypotheses.

Activity

Find a paper in your field that reports "no significant difference". Locate the confidence interval. Decide whether the study showed there was no meaningful effect, or simply could not tell — and write one sentence explaining how the interval settles it.

Further reading

- **Greenland et al. (2016)** — "Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations" — Open access. Twenty-five common misreadings, each corrected. Essential.
- **Wasserstein & Lazar (2016)** — **ASA Statement on p-Values** — The profession's own statement of what p-values cannot do.
- **Sullivan & Feinn (2012)** — "Using Effect Size — or Why the P Value Is Not Enough" — Short and practical.

Self-check questions

1. A 95% confidence interval for a mean difference is 2.1 to 8.7 mmHg. Which statement is correct?
 - a. There is a 95% probability the true difference lies between 2.1 and 8.7
 - b. If the study were repeated many times, 95% of such intervals would contain the true difference
 - c. 95% of participants had a difference in that range
 - d. The result is not statistically significant
2. $p = 0.03$ means:
 - a. There is a 3% probability the null hypothesis is true
 - b. If the null were true, data at least this extreme would occur 3% of the time
 - c. There is a 97% probability the finding is real
 - d. The effect is small
3. A study of 50,000 people finds a mean difference of 0.4 mmHg with $p < 0.001$. The most sensible reading is:
 - a. A clinically important effect has been demonstrated
 - b. The effect is statistically detectable but far too small to matter clinically
 - c. The analysis must be wrong
 - d. The sample was too small

4. Testing 20 independent hypotheses at the 5% level gives roughly what chance of at least one false positive?
- 5%
 - 20%
 - 64%
 - 95%
5. A trial with 30% power finds no significant difference. You should conclude:
- The treatment does not work
 - The study was too small to detect an effect of the size expected, so it is uninformative
 - The null hypothesis is proven
 - The p-value must be wrong
6. Reporting "p = 0.000" is poor practice because:
- p-values cannot be reported to three decimals
 - A p-value is never exactly zero; it should be written $p < 0.001$
 - It implies the sample was too large
 - SPSS never produces that output
-

Answers

- b.** The confidence statement is about the procedure across repeated sampling, not about this one interval. The true value is fixed; the interval is what varies. Option C confuses a confidence interval with a reference range.
 - b.** The p-value is computed assuming the null is true, so it cannot be the probability that the null is true. It is a statement about the data given a hypothesis, not about a hypothesis given the data.
 - b.** Large samples make trivial differences statistically significant. 0.4 mmHg is well inside measurement error and would change no decision. This is exactly why effect sizes must be reported alongside p-values.
 - c.** $1 - 0.95^{20}$ is about 0.64. Running many tests and reporting the significant ones is why pre-specifying a primary outcome, or adjusting for multiplicity, matters.
 - b.** With 30% power the study would miss a real effect 70% of the time. Absence of evidence here is not evidence of absence; the confidence interval will be wide and consistent with an important effect.
 - b.** Software rounds a very small value to 0.000. The probability is small, not zero, so the correct notation is $p < 0.001$.
-

Comparing Groups

180 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Choosing the right test for your design, outcome and assumptions — t-tests, ANOVA, chi-square and their non-parametric alternatives — and running them in SPSS and R.

Learning objectives

- Select a test from the design, the outcome type and the number of groups
 - Check the assumptions each test relies on
 - Distinguish paired from independent comparisons
 - Run and interpret t-tests, ANOVA with post-hoc tests, and chi-square
 - Choose a non-parametric alternative when assumptions fail
 - Report a comparison completely: estimate, interval, test statistic and effect size
-

Choosing the test

- Three questions settle it: what type is the outcome, how many groups, and are the groups independent or paired?
- Continuous outcome, two independent groups: independent samples t-test
- Continuous outcome, two paired measurements: paired t-test
- Continuous outcome, three or more independent groups: one-way ANOVA
- Categorical outcome, two or more groups: chi-square test of independence
- Small expected counts in a contingency table: Fisher's exact test
- Paired binary outcome: McNemar's test

TEST SELECTION

Outcome	Groups	Parametric	Non-parametric
Continuous	2 independent	Independent t-test	Mann-Whitney U
Continuous	2 paired	Paired t-test	Wilcoxon signed-rank
Continuous	3+ independent	One-way ANOVA	Kruskal-Wallis
Continuous	3+ repeated	Repeated-measures ANOVA	Friedman
Categorical	2+ independent	Chi-square	Fisher's exact
Binary	2 paired	McNemar's test	—

Assumptions, and what to do when they fail

- Independence of observations — the one assumption no test survives losing, and no fix repairs after the fact
- Approximate normality of the outcome within groups; matters most in small samples
- Equal variances for the classic t-test — but Welch's t-test drops this and is a sensible default
- For chi-square, expected counts of at least 5 in most cells
- Judge normality from a histogram, not only a significance test
- When assumptions fail: transform the outcome, use a non-parametric test, or use a bootstrap
- Non-parametric tests compare distributions or ranks, not means, so interpret accordingly

IN PRACTICE

Clustered data — pupils within schools, patients within clinics, repeated visits per person — violate independence. A t-test on such data reports standard errors that are far too small. That needs a mixed model or a cluster-robust approach, not a different t-test.

ANOVA and post-hoc tests

- A significant ANOVA says the groups are not all equal; it does not say which differ
- Running every pairwise t-test afterwards without adjustment inflates the false positive rate
- Post-hoc tests control it: Tukey HSD for all pairwise comparisons, Dunnett when comparing several groups against one control
- Bonferroni is simple and conservative; it divides the significance level by the number of comparisons
- Two-way ANOVA tests two factors and their interaction — an interaction means the effect of one factor depends on the level of the other
- Interpret main effects with care in the presence of a significant interaction

Running it

- R, independent t-test: `t.test(bp ~ group, data = dat)` — Welch by default
- R, paired: `t.test(before, after, paired = TRUE)`
- R, ANOVA: `fit <- aov(bp ~ group, data = dat); summary(fit); TukeyHSD(fit)`
- R, chi-square: `chisq.test(table(dat$group, dat$outcome))`
- SPSS: Analyze > Compare Means > Independent-Samples T Test, or One-Way ANOVA with Post Hoc
- SPSS chi-square lives under Analyze > Descriptive Statistics > Crosstabs > Statistics
- In SPSS always read the Levene test row to decide which t-test line to quote

Reporting a comparison

- Give the group summaries first: mean (SD) or median (IQR) with n per group
- Give the estimated difference with its confidence interval — the interval is the finding
- Give the test statistic, degrees of freedom and exact p-value
- Give an effect size
- Example: "Mean systolic BP was 8.4 mmHg lower in the intervention group (95% CI 3.1 to 13.7; $t(118) = 3.12$, $p = 0.002$; $d = 0.57$)."
- Name the test used and state that its assumptions were checked

Worked examples

Choosing a test in three questions

Four analyses are needed, and the student wants to know which test each requires.

1. Mean BP in intervention vs control, different people: continuous, two groups, independent → independent t-test (Welch).
2. Anxiety score before and after in the same patients: continuous, two measurements, paired → paired t-test.
3. Mean score across four clinics: continuous, four groups, independent → one-way ANOVA, then Tukey.
4. Proportion adhering in two groups: categorical outcome, two groups → chi-square, or Fisher if expected counts are small.
5. Each answer came from three questions: outcome type, number of groups, paired or independent.
6. Only then check assumptions and consider a non-parametric alternative.

THE POINT

Test choice is a short decision tree, not memorisation. Getting the paired versus independent question wrong is the most common error, and it changes both the test and the power.

Pupils in schools are not independent

Test scores from 600 pupils across 20 schools are compared between two teaching methods with an independent *t*-test. The result is $p = 0.004$.

1. Pupils in the same school share teachers, resources and intake — their scores are correlated.
2. A *t*-test assumes 600 independent observations. In effect there are closer to 20.
3. Treating correlated observations as independent understates the standard error, so p is far too small.
4. The design effect quantifies it: with an intra-class correlation of 0.15 and 30 pupils per school, variance inflates by about 5.
5. The right analysis is a mixed model with a random intercept for school: `lmer(score ~ method + (1 | school), data = dat)`
6. Refitted properly, the same data give $p = 0.14$ — the original finding was an artefact of ignoring clustering.

THE POINT

Independence is the one assumption no test survives losing, and no transformation repairs. Whenever observations are grouped — schools, clinics, villages, repeated visits — the clustering must be in the model.

Key terms

Welch's *t*-test

A *t*-test that does not assume equal variances; a safe default for two independent groups.

Paired test

A comparison of two measurements on the same units, analysing the within-pair differences.

Post-hoc test

A follow-up comparison after a significant ANOVA, adjusted for multiple testing.

Interaction

When the effect of one factor depends on the level of another.

Fisher's exact test

An exact test for a contingency table, used when expected counts are small.

Cohen's *d*

A standardised difference between two means, expressed in pooled standard deviations.

Clustering

Non-independence arising when observations are grouped within units such as schools or clinics.

Activity

For three research questions from your own study, write down: the outcome type, the number of groups, whether they are paired, and the test you would use. Then name the one assumption most likely to fail in each, and what you would do about it.

Further reading

- **Altman & Bland** — "Parametric v non-parametric methods" — BMJ Statistics Notes, one page, settles most arguments.
- **Delacre, Lakens & Leys (2017)** — "Why Psychologists Should by Default Use Welch's t-test" — The case for dropping the equal-variance assumption.
- **Field** — **Discovering Statistics Using IBM SPSS Statistics** — Step-by-step SPSS with output interpretation.

Self-check questions

1. You measure anxiety in the same 40 patients before and after a programme. The appropriate test is:
 - a. Independent samples t-test
 - b. Paired t-test
 - c. One-way ANOVA
 - d. Chi-square
2. A one-way ANOVA across four groups gives $p = 0.004$. You may conclude:
 - a. All four groups differ from each other
 - b. The groups are not all equal, but a post-hoc test is needed to say which differ
 - c. Group 1 differs from group 4
 - d. The largest and smallest means are significantly different
3. A 2×2 table has expected counts of 3, 4, 12 and 15. The appropriate test is:
 - a. Chi-square
 - b. Fisher's exact test
 - c. Independent t-test
 - d. Kruskal-Wallis
4. You analyse test scores from 600 pupils across 20 schools with an ordinary t-test. The main problem is that:

- a. The sample is too large
 - b. Pupils within a school are not independent, so standard errors will be too small
 - c. Test scores cannot be compared
 - d. A t-test cannot handle 600 observations
5. Levene's test in your SPSS output is significant. This tells you:
- a. The data are not normally distributed
 - b. The group variances differ, so the equal-variances-not-assumed (Welch) row should be read
 - c. The result is not significant
 - d. You must use a chi-square test
6. Which reported result is complete?
- a. The difference was significant ($p < 0.05$).
 - b. Mean BP was 8.4 mmHg lower in the intervention group (95% CI 3.1 to 13.7; $t(118) = 3.12$, $p = 0.002$; $d = 0.57$).
 - c. There was a significant effect of the intervention on blood pressure.
 - d. ANOVA showed $p = 0.002$.
-

Answers

1. **b.** The two measurements come from the same people, so they are paired. A paired t-test analyses the within-person differences, which removes between-person variation and is more powerful than treating the groups as independent.
 2. **b.** ANOVA is an omnibus test. It rejects the hypothesis that all means are equal without identifying where the differences lie — that requires an adjusted post-hoc comparison such as Tukey.
 3. **b.** Chi-square relies on an approximation that becomes unreliable when expected counts fall below about 5. Fisher's exact test computes the probability directly and has no such requirement.
 4. **b.** Pupils in the same school share teachers, intake and environment. Ignoring that clustering treats correlated observations as independent, understating standard errors and overstating significance. A mixed model or cluster-robust errors are needed.
 5. **b.** Levene tests homogeneity of variance, not normality. When it is significant, quote the Welch row. Many statisticians now recommend Welch's t-test as the default regardless.
 6. **b.** A complete report gives the estimate, its confidence interval, the test statistic with degrees of freedom, the exact p-value and an effect size. The others tell the reader almost nothing about magnitude or precision.
-

Correlation and Linear Regression

180 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Modelling a continuous outcome: what a regression coefficient means, how to handle categorical predictors, and how to check the model is entitled to the conclusions you want from it.

Learning objectives

- Distinguish correlation from regression and know when each is appropriate
- Interpret a slope, an intercept and R-squared in the units of the problem
- Include categorical predictors correctly using dummy coding
- Extend to multiple regression and interpret adjusted coefficients
- Check the assumptions using residual plots
- Report a regression model completely

Correlation

- Pearson r measures the strength and direction of a linear relationship between two continuous variables
- It runs from -1 to $+1$; 0 means no linear relationship, not no relationship at all
- Spearman ρ is the rank-based alternative for monotonic but non-linear relationships, or with outliers
- r -squared is the proportion of variance in one variable explained by the other
- Always plot the scatterplot before quoting r — very different patterns give identical coefficients
- Correlation is symmetric and makes no prediction; regression is directional and does

Simple linear regression

- The model is $y = b_0 + b_1 \cdot x + \text{error}$
- b_1 is the estimated change in y for a one-unit increase in x
- b_0 is the predicted y when x is zero — often meaningless, as with a predicted birth weight at age zero
- Centring a predictor (subtracting its mean) makes the intercept interpretable as the value at the average
- R-squared is the proportion of variance explained; a low value does not mean the model is wrong, only that much is unexplained
- In R: `fit <- lm(bp ~ age, data = dat); summary(fit); confint(fit)`
- Always report the coefficient with its confidence interval, not just its p-value

Categorical predictors

- A binary predictor enters as 0/1, and its coefficient is the difference between the two groups
- A k-category predictor needs k-1 dummy variables, with one level as the reference
- Every coefficient is then interpreted as the difference from that reference level
- Choose a reference that makes sense — usually the largest group, or the untreated one
- In R declare it a factor and `lm()` handles the coding: `dat$edu <- factor(dat$edu)`
- Entering an unordered categorical variable as a plain number is a common and serious error: it forces a straight-line effect across arbitrary category codes

IN PRACTICE

Coding education as 1 to 5 and putting it in as numeric assumes the step from none to primary equals the step from SHS to tertiary, and that both are one unit of the same thing. Almost never true — use factors.

Multiple regression

- Each coefficient is the effect of that predictor holding the others constant
- "Adjusted for" and "holding constant" mean the same thing here
- Adding a confounder changes the coefficient of interest; that change is the confounding
- Adjusted R-squared penalises adding predictors that do not help
- Do not select variables purely by p-value; stepwise selection produces unstable models and biased intervals
- Choose predictors from subject knowledge and the conceptual framework built in Part 1
- A rough guide: at least 10-15 observations per predictor, more for binary outcomes

READING A REGRESSION TABLE

Predictor	b (95% CI)	p	Interpretation
Age (years)	0.42 (0.28 to 0.56)	<0.001	Each extra year adds 0.42 mmHg
Sex (female vs male)	-3.10 (-5.9 to -0.3)	0.030	Women 3.1 mmHg lower than men
BMI (kg/m ²)	0.88 (0.55 to 1.21)	<0.001	Each BMI unit adds 0.88 mmHg
Education: tertiary vs none	-2.40 (-5.1 to 0.3)	0.081	Lower, but interval includes no difference

Checking the model

- Linearity: residuals versus fitted values should show no pattern; a curve means the relationship is not linear
- Constant variance: the same plot should not fan out — a funnel shape is heteroscedasticity
- Normality of residuals: a Q-Q plot should follow the diagonal. It is the residuals that matter, not the raw data
- Independence: cannot be checked from residuals alone; it follows from the design
- Influential points: Cook's distance identifies observations that move the coefficients substantially
- Multicollinearity: a variance inflation factor above about 5 to 10 means predictors are too closely related to separate
- In R: `plot(fit)` gives the four diagnostic plots; `vif(fit)` from the `car` package gives VIFs

Worked examples

Reading a regression table out loud

A model of systolic blood pressure gives: age 0.42 (0.28 to 0.56); female -3.10 (-5.9 to -0.3); BMI 0.88 (0.55 to 1.21); tertiary education -2.40 (-5.1 to 0.3).

1. Age: each additional year is associated with 0.42 mmHg higher SBP, holding sex, BMI and education constant.
2. Female: women average 3.1 mmHg lower than men, adjusted for the others. The reference category is male.
3. BMI: each additional kg/m² adds 0.88 mmHg — over 5 units, about 4.4 mmHg.
4. Tertiary education: 2.4 mmHg lower than the reference level, but the interval crosses zero, so this is compatible with no difference.
5. Every statement needs "holding the others constant" — that is what adjustment means.
6. Reporting only "age was significant" throws away the magnitude, which is the part a clinician can use.

THE POINT

A coefficient is a sentence in the units of the problem, not a p-value. If you cannot say it out loud in those units, you cannot yet interpret your own model.

The education variable entered as a number

Education has five categories coded 1 to 5. It is entered into the regression as numeric and returns a neat coefficient of -1.2 per level.

1. That single coefficient assumes each step has the same effect: none→primary equals SHS→tertiary.
2. It also assumes the categories are equally spaced on some underlying scale, which nobody has established.
3. Declaring it a factor — `dat$edu <- factor(dat$edu)` — estimates four coefficients against a reference.
4. Refitted, the effect turns out to be flat across the first three levels and then drops sharply at tertiary.
5. The linear version averaged that into a misleading constant slope and understated the tertiary effect.
6. If a single-number summary is genuinely wanted, test for trend explicitly and say that is what you did.

THE POINT

Numeric coding of an unordered or unequally spaced categorical variable imposes a straight line nobody checked. Use factors, and let the data show the shape.

Key terms

Pearson r

A measure of linear association between two continuous variables, from -1 to +1.

Regression coefficient

The estimated change in the outcome per one-unit increase in a predictor.

Dummy variable

A binary indicator representing one level of a categorical predictor against a reference.

Adjusted

Estimated while holding the other predictors in the model constant.

Heteroscedasticity

Non-constant variance of residuals across fitted values.

Multicollinearity

Predictors so correlated that their separate effects cannot be estimated reliably.

Cook's distance

A measure of how much an observation influences the fitted coefficients.

Activity

Fit a regression with one continuous and one categorical predictor on your own data or a supplied dataset. Write one sentence interpreting each coefficient in the units of the problem, then run the four diagnostic plots and say which assumption you are least comfortable with.

Further reading

- **Vittinghoff et al. — Regression Methods in Biostatistics** — The standard applied text; strong on confounding and model building.
- **Altman & Bland — "Correlation, regression and repeated data"** — BMJ Statistics Notes, short and corrective.
- **Harrell — Regression Modeling Strategies** — Advanced; definitive on why stepwise selection should be avoided.

Self-check questions

1. In a regression of blood pressure on age, the coefficient for age is 0.42. This means:
 - a. 42% of blood pressure is explained by age
 - b. Each additional year of age is associated with a 0.42 mmHg higher blood pressure
 - c. Age and blood pressure correlate at $r = 0.42$
 - d. Blood pressure rises 0.42% per year
2. Education has five ordered categories. Entering it as a numeric variable 1-5 assumes:
 - a. Nothing problematic
 - b. That each step between adjacent categories has the same effect on the outcome
 - c. That education is normally distributed
 - d. That the reference category is "none"
3. A residuals-versus-fitted plot shows a clear funnel widening to the right. This indicates:
 - a. Non-normal residuals
 - b. Heteroscedasticity — the variance is not constant across fitted values
 - c. Multicollinearity
 - d. An omitted interaction
4. Adding a confounder to a model changes the coefficient of interest from 5.2 to 2.1. This means:
 - a. The model is misspecified

- b. Much of the crude association was explained by that confounder
 - c. The confounder should be removed
 - d. The sample is too small
 - 5. A Q-Q plot in regression diagnostics assesses the normality of:
 - a. The outcome variable
 - b. The residuals
 - c. The predictors
 - d. The fitted values
 - 6. Stepwise selection by p-value is discouraged mainly because it:
 - a. Is computationally slow
 - b. Produces unstable models with biased coefficients and confidence intervals that are too narrow
 - c. Cannot handle categorical predictors
 - d. Requires very large samples
-

Answers

1. **b.** A regression coefficient is the change in the outcome per one-unit change in the predictor, in the outcome's own units. Variance explained is R-squared; correlation is r.
 2. **b.** Numeric entry imposes a constant linear effect per category step. Declaring it a factor estimates a separate coefficient for each level against a reference, which is almost always the honest choice.
 3. **b.** A funnel means spread changes with the fitted value. Coefficients stay unbiased but standard errors are wrong; a transformation or robust standard errors is the usual fix.
 4. **b.** That change is exactly what confounding looks like. The adjusted estimate is the effect holding the confounder constant, and reporting both crude and adjusted lets the reader see how much it mattered.
 5. **b.** Linear regression assumes normally distributed errors, not normally distributed raw data. A skewed outcome can still yield well-behaved residuals once predictors are in the model.
 6. **b.** Letting the data choose the model and then reporting intervals as if it had been pre-specified understates uncertainty, and small changes in the data can produce a different model. Choose predictors from subject knowledge instead.
-

Logistic Regression and Confounding

180 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Modelling binary outcomes: odds, odds ratios and what they do and do not mean, plus adjusting for confounding and recognising mediators and interactions.

Learning objectives

- Explain why linear regression is unsuitable for a binary outcome
- Interpret odds ratios and their confidence intervals correctly
- Distinguish odds ratio from risk ratio and know when they diverge
- Fit and interpret multivariable logistic regression
- Distinguish confounders, mediators and colliders, and adjust accordingly
- Assess model fit and predictive performance

Why not linear regression

- A binary outcome is 0 or 1, but a linear model can predict below 0 and above 1
- The residuals cannot be normal or homoscedastic when the outcome takes two values
- Logistic regression models the log odds, which is unbounded and well behaved
- The fitted probability is then obtained by transforming back through the logistic function
- Odds are $p / (1 - p)$: a probability of 0.2 corresponds to odds of 0.25
- The model is linear in the log odds, not in the probability — which is why effects are multiplicative

Reading an odds ratio

- $\exp(b)$ is the odds ratio for a one-unit increase in the predictor
- $OR = 1$ means no association; above 1 higher odds; below 1 lower odds
- The confidence interval is what matters: an OR of 3.2 with an interval of 0.8 to 12.6 is not evidence of much
- Odds ratios are not risk ratios. When the outcome is rare (under about 10%) they are close; when it is common they diverge sharply
- An OR of 3 for an outcome affecting 40% of people does not mean three times the risk

- Report the interval always, and consider a risk ratio or risk difference where the outcome is common and the design allows it
- In R: `fit <- glm(y ~ x, family = binomial, data = dat); exp(cbind(OR = coef(fit), confint(fit)))`

IN PRACTICE

The odds ratio is the coefficient logistic regression naturally produces, and it is often reported as though it were a risk ratio. For a common outcome that overstates the effect substantially — a mistake that appears in print constantly.

Confounders, mediators and colliders

- A confounder causes both the exposure and the outcome; adjusting for it removes bias
- A mediator lies on the causal path from exposure to outcome; adjusting for it removes part of the effect you are trying to measure
- A collider is caused by both exposure and outcome; adjusting for it creates bias where none existed
- The distinction cannot be made from the data — it comes from subject knowledge and the conceptual framework
- Draw a directed acyclic graph before choosing what to adjust for
- Adjusting for everything available is not conservative; it can actively introduce bias
- Example: adjusting smoking for lung function when estimating smoking's effect on mortality removes part of the very pathway of interest

WHAT TO DO WITH A THIRD VARIABLE

Role	Relationship	Adjust?	Effect of adjusting
Confounder	Causes exposure and outcome	Yes	Removes bias
Mediator	On the path exposure → outcome	Usually no	Removes part of the effect
Collider	Caused by exposure and outcome	No	Creates bias
Competing cause	Causes outcome only	Optional	Improves precision

Building and checking the model

- Rule of thumb: at least 10 events per predictor, counting the rarer outcome category, not the sample size
- 40 events and 8 predictors is over-fitted regardless of how large the sample looks
- Check linearity of continuous predictors on the log-odds scale, not the probability scale
- Hosmer-Lemeshow tests calibration, though it is sensitive to sample size and grouping
- The C-statistic (area under the ROC curve) measures discrimination: 0.5 is chance, above 0.8 is good

- Report both discrimination and calibration for a prediction model; either alone is incomplete
- Check for separation — a predictor that perfectly predicts the outcome produces enormous coefficients and unusable intervals

Reporting

- Report the number of events and the number of observations, not only n
- Present crude and adjusted estimates side by side
- State which variables were adjusted for and why they were chosen
- Give odds ratios with confidence intervals to two decimal places
- Say whether continuous predictors were entered linearly, categorised, or modelled with splines
- If the outcome is common, note that the odds ratio overstates the risk ratio

Worked examples

An odds ratio read as a risk ratio

A study of a common outcome — 40% of the unexposed group — reports $OR = 3.0$. The abstract says the exposure "triples the risk".

1. Odds in the unexposed: $0.40 / 0.60 = 0.67$.
2. Odds in the exposed: $0.67 \times 3.0 = 2.0$, so probability = $2.0 / 3.0 = 0.67$.
3. So risk goes from 40% to 67% — a risk ratio of about 1.7, not 3.0.
4. The odds ratio and risk ratio converge only when the outcome is rare, under roughly 10%.
5. At a 40% baseline they diverge substantially, and the abstract overstates the effect by nearly double.
6. Where the design allows, report a risk ratio or risk difference for a common outcome.

THE POINT

Logistic regression naturally produces odds ratios, and they are routinely described as risks. For common outcomes that is a real and large overstatement, and it appears in print constantly.

Adjusting for a mediator by accident

A study of smoking and mortality adjusts for age, sex, income and lung function. The smoking effect shrinks sharply, and the authors conclude smoking matters less than thought.

1. Age, sex and income plausibly cause both smoking and mortality — they are confounders, and adjusting is right.
2. Lung function is different: smoking damages it, and that damage causes mortality.
3. So lung function lies on the causal path from exposure to outcome — it is a mediator.
4. Adjusting for it removes the part of smoking's effect that operates through lung damage, which is most of it.
5. The adjusted estimate answers "the effect of smoking not acting through lung function", which is not the question asked.
6. Drawing a causal diagram before choosing covariates would have separated the confounders from the mediator.

THE POINT

Adjusting for everything available is not the cautious choice. Which variables to adjust for is a causal question answered from subject knowledge, not a statistical one answered from the data.

Key terms

Odds

The probability of an event divided by the probability of it not occurring.

Odds ratio

The ratio of odds between two groups; the natural output of logistic regression.

Risk ratio

The ratio of probabilities between two groups; diverges from the odds ratio when the outcome is common.

Confounder

A variable causing both exposure and outcome, biasing the crude association.

Mediator

A variable on the causal path between exposure and outcome.

Collider

A variable caused by both exposure and outcome; adjusting for it introduces bias.

C-statistic

Area under the ROC curve, measuring how well a model discriminates between outcomes.

Events per variable

The number of outcome events divided by the number of predictors, a guide to over-fitting.

Activity

For your own binary outcome, list every variable you might adjust for. Classify each as confounder, mediator, collider or competing cause, and justify the classification from what causes what — not from what the data show. Then state which you will adjust for.

Further reading

- **Hernán & Robins — Causal Inference: What If** — Free online. The clearest treatment of confounders, mediators and colliders.
- **Knol et al. (2012) — "What do case-control studies estimate?"** — On odds ratios versus risk ratios.
- **Peduzzi et al. (1996) — events per variable in logistic regression** — The origin of the ten-events-per-predictor guide.

Self-check questions

1. A study reports $OR = 3.0$ for an outcome occurring in 40% of the unexposed group. Reading this as "three times the risk" is:
 - a. Correct
 - b. An overstatement, because odds ratios exceed risk ratios when the outcome is common
 - c. An understatement
 - d. Correct only in case-control studies
2. You are estimating the effect of smoking on mortality and adjust for lung function. Lung function is most likely a:
 - a. Confounder, so adjustment is correct
 - b. Mediator, so adjusting removes part of the effect you are trying to measure
 - c. Collider
 - d. Competing cause
3. Your dataset has 500 participants, 40 of whom had the outcome, and you fit 8 predictors. The concern is:
 - a. The sample is too small overall
 - b. Only 5 events per predictor, so the model is over-fitted
 - c. Logistic regression cannot handle 8 predictors
 - d. The outcome is too common

4. An odds ratio of 2.4 with a 95% CI of 0.7 to 8.1 should be described as:
- A strong positive association
 - Consistent with anything from a modest protective effect to a large harmful one — the study is uninformative here
 - Evidence of no association
 - Statistically significant
5. A C-statistic of 0.52 indicates the model:
- Discriminates well
 - Discriminates barely better than chance
 - Is perfectly calibrated
 - Has too many predictors
6. Which is the correct R call for logistic regression?
- `lm(y ~ x, data = dat)`
 - `glm(y ~ x, family = binomial, data = dat)`
 - `aov(y ~ x, data = dat)`
 - `cor.test(y, x)`
-

Answers

- b.** Odds and risk diverge as the outcome becomes common. With a 40% baseline, an OR of 3.0 corresponds to a risk ratio well under 2. The approximation only holds for rare outcomes.
 - b.** Smoking damages lung function, which contributes to mortality — it lies on the causal path. Adjusting for it answers a different question: the effect of smoking not acting through lung function.
 - b.** What limits a logistic model is the number of events, not the sample size. 40 events with 8 predictors is 5 per variable, below the usual guide of 10, giving unstable coefficients and intervals that are too narrow.
 - b.** The interval spans 1 and is very wide. The point estimate is not the finding; the interval is, and here it tells you the study could not pin the effect down.
 - b.** 0.5 is a coin flip and 1.0 is perfect discrimination. 0.52 means the model can hardly separate those who had the outcome from those who did not, whatever its p-values say.
 - b.** `glm` with `family = binomial` fits the logistic model. Omitting the family argument fits a linear model instead — which runs without error and gives quietly wrong results.
-

Reporting, Reproducibility and the Results Section

150 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Turning output into a results section a reviewer will accept: what to report, how to lay out tables, and how to make the whole analysis reproducible by someone else.

Learning objectives

- Write a results section that matches the pre-specified analysis plan
 - Report estimates, intervals and p-values to the right precision
 - Lay out tables and figures to journal standards
 - Apply the relevant reporting checklist
 - Build a reproducible analysis with a script and a clear folder structure
 - Produce a dynamic report with R Markdown or Quarto
-

What belongs in the results

- Participant flow: how many were approached, included, excluded and analysed, and why
 - Table 1 of participant characteristics
 - The primary analysis, answering the primary objective, first
 - Secondary analyses clearly labelled as secondary
 - Any analysis not planned in advance labelled exploratory — not doing so is the difference between honest reporting and fishing
 - No interpretation in the results section; that belongs in the discussion
 - Every objective from the proposal produces a result, and every result answers an objective
-

Precision and notation

- Report to the precision the data support: a mean age of 42.3 years, not 42.3187
- Percentages to at most one decimal, and never for a denominator under about 20
- p-values to two or three decimals; below 0.001 write $p < 0.001$, never $p = 0.000$
- Effect estimates with confidence intervals, using "to" rather than a dash when values may be negative
- Give the unit of measurement in the row or column label, not repeated in every cell

- Be consistent about decimal places within a column so figures line up

COMMON REPORTING FAULTS

Written	Problem	Better
$p = 0.000$	No p-value is zero	$p < 0.001$
Mean 42.3187 years	False precision	Mean 42.3 years
OR 2.4 ($p < 0.05$)	No interval	OR 2.4 (95% CI 1.3 to 4.5)
33.3% ($n = 3$)	Percentage of a tiny denominator	1 of 3
A trend towards significance	$p = 0.08$ is not a trend	No significant difference ($p = 0.08$)

Checklists

- STROBE for observational studies — cohort, case-control and cross-sectional
- CONSORT for randomised trials
- PRISMA for systematic reviews
- TRIPOD for prediction model studies
- SAMPL for statistical reporting generally
- Find them at the EQUATOR Network, and use one while writing rather than afterwards
- Most journals require the completed checklist at submission anyway

Making the analysis reproducible

- A reproducible analysis means someone else can run your script on your data and get your numbers
- Structure: data/raw (read-only), data/clean, scripts, output, and a README explaining the order
- Number scripts in running order: 01-clean.R, 02-describe.R, 03-model.R
- Never edit data by hand; every change is a line of code
- Set a seed before anything random so imputation and bootstrap results repeat exactly
- Record package versions — `sessionInfo()` in R, or `renv` for a full lockfile
- Use relative paths so the project runs on another machine

IN PRACTICE

The practical test: delete every output file, rerun the scripts from raw data, and check the numbers in your manuscript still match. Most analyses fail this the first time, which is exactly why it is worth doing before a reviewer asks.

Dynamic reports

- R Markdown and Quarto embed the code that produces each number in the document itself
- The manuscript is then generated from the data, so a table can never disagree with the analysis
- Inline code writes results into prose: the mean was ``r round(mean(dat$age), 1)`` years
- Re-running after a data correction updates every number, table and figure at once
- Output to Word for co-authors, PDF for submission, or HTML for sharing
- This removes the single most common source of error in a results section: copying a number by hand and later changing the analysis

Worked examples

A results paragraph rewritten

A draft reads: "The intervention group had significantly lower blood pressure ($p < 0.05$), showing that the programme works well and should be rolled out nationally."

1. No estimate: how much lower? The reader cannot tell if it is 1 mmHg or 20.
2. No interval: no sense of precision.
3. $p < 0.05$ rather than the exact value, which discards information.
4. "Showing that the programme works well" is interpretation, and belongs in the discussion.
5. "Should be rolled out nationally" is a recommendation, and belongs in the conclusion — supported by more than one trial.
6. Rewritten: "Mean systolic BP was 8.4 mmHg lower in the intervention group (95% CI 3.1 to 13.7; $t(118) = 3.12$, $p = 0.002$; $d = 0.57$)."

THE POINT

The results section reports what was found, in numbers, with uncertainty. Every clause of interpretation that creeps in is a clause a reviewer will ask you to move.

The table that no longer matched the analysis

Two weeks before submission a data error is found and corrected. The analysis is rerun. The manuscript is submitted, and a co-author notices Table 3 still carries the old numbers.

1. The tables had been typed by hand from earlier output.
2. Rerunning the analysis updated the output files but not the manuscript.
3. Three of eleven numbers in Table 3 changed; two did not, so nothing looked obviously wrong.
4. In Quarto or R Markdown the table is generated from the data at render time, so this cannot happen.
5. Inline code puts results into prose too: the mean was ``r round(mean(dat$age), 1)`` years.
6. Re-rendering after the correction would have updated every number, table and figure at once.

THE POINT

Copying numbers by hand and later changing the analysis is the most common source of error in a results section. Generating the document from the data makes the failure impossible rather than unlikely.

Key terms

Primary analysis

The pre-specified analysis answering the study's main objective.

Exploratory analysis

An analysis not planned in advance, which must be labelled as such.

STROBE

The reporting checklist for observational studies.

Reproducibility

The property that another person can obtain your results from your data and code.

Seed

A starting value for the random number generator, making stochastic procedures repeatable.

Dynamic report

A document in which results are generated from the data at render time rather than pasted in.

Activity

Take one table from a draft of your own results. Check it against the reporting faults table in this unit, fix every instance, and add a confidence interval to every estimate that lacks one. Then write the accompanying paragraph without a single word of interpretation.

Further reading

- **EQUATOR Network — reporting guidelines** — Free index of every checklist. Find yours before you write.
- **Lang & Altman — SAMPL guidelines for statistical reporting** — Concrete rules for numbers, tables and notation.
- **Quarto documentation** — Works with R, Python and Stata; the successor to R Markdown.
- **The Turing Way — Guide for Reproducible Research** — Free, community-written, practical on project structure.

Self-check questions

1. Your software reports $p = 0.000$. You should write:
 - a. $p = 0.000$
 - b. $p < 0.001$
 - c. $p = 0$
 - d. p is highly significant
2. You ran an unplanned subgroup analysis that produced an interesting finding. In the paper it must be:
 - a. Reported as a primary result
 - b. Labelled exploratory, with the reader told it was not pre-specified
 - c. Omitted entirely
 - d. Reported without a p -value
3. "There was a trend towards significance ($p = 0.08$)" is poor practice because:
 - a. $p = 0.08$ is not reportable
 - b. A p -value is not a trend; the result either meets the threshold or does not, and the interval is what conveys the uncertainty
 - c. Trends require at least three time points
 - d. The threshold should have been 0.10
4. The practical test of a reproducible analysis is that:
 - a. The code is commented
 - b. Deleting all outputs and rerunning the scripts from raw data reproduces every number in the manuscript
 - c. It was run in R rather than SPSS
 - d. The data are publicly available

5. Setting a random seed before multiple imputation matters because:
- a. It makes the imputation more accurate
 - b. It makes the results exactly repeatable on a rerun
 - c. It is required by Rubin's rules
 - d. It reduces the number of imputations needed
6. The main advantage of writing results in Quarto or R Markdown is that:
- a. It produces prettier documents
 - b. Numbers are generated from the data at render time, so a table can never disagree with the analysis
 - c. It runs the analysis faster
 - d. It removes the need for a statistician
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Answers

- 1. **b.** The value has been rounded, not computed as zero. A probability of exactly zero would mean the data are impossible under the null, which is never the case.
 - 2. **b.** Unplanned analyses are legitimate and often valuable, but presenting one as though it had been planned misrepresents the false positive risk. Label it and let the reader weigh it accordingly.
 - 3. **b.** The phrase implies the effect would have been significant with more data, which is not something the p-value says. Report the estimate and interval and let them speak.
 - 4. **b.** Reproducibility is demonstrated, not asserted. Rerunning from raw data is the only check that catches hand-edited cells, undocumented steps and numbers that drifted after the analysis changed.
 - 5. **b.** Imputation is stochastic, so results differ slightly each run. A seed fixes the random sequence so you and a reviewer obtain identical numbers. It does not affect validity.
 - 6. **b.** Copying numbers by hand and then changing the analysis is the commonest source of inconsistency in a results section. Generating them from the data at render time makes that failure impossible.
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